

Long Questions Home-Work (Periodic Table)

1. Henry Mosely in 1913 classified the elements based on Modern periodic law.

i. State Modern periodic law.

→ The modern periodic law states that the physical and chemical property of an element is a periodic function of their atomic numbers.

ii. What is periodicity?

→ The repetition of similar properties of elements after certain intervals when arranged according to atomic number is called periodicity.

iii. How does modern periodic table explain the cause of periodicity?

→ The modern periodic table explains the cause of periodicity as follows:-

i. Repetition of similar valence shell configuration

iv. Write the electronic configuration of an element having atomic number 24.

→ Electronic configuration of element having atomic number 24:
 $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1, 3d^5$

v. What type of bond is formed between an element 'A' having valence electronic configuration $3s^1$ & element 'B' having valence shell electronic configuration $3s^2, 3p^5$.

→ In element 'A' there only 1 valence e^- so it must be a metal of group IA and (Na), likewise in element 'B' there are 7 valence e^- so it must be a halogen (Cl). So an ionic bond is formed between A and B.

2. An element has atomic number 26.

i. Write its electronic configuration.

→ electronic configuration of element with atomic number 26.
 $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^6$

ii. Predict its block type & write two important properties of the same block type.

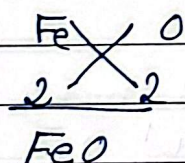
→ As the outermost e^- lies in the d-block the element lies in block d. Some of its important properties are:-

1. D-block elements can ~~pair out~~ be attracted by magnet due to free electrons.

2. These elements can have variable valency.

iii. Which compound is formed when the above element in the +2 oxidation state is combined with the element Y having valence shell electronic configuration $2s^2, 2p^4$.

→ The element with atomic number 26 is iron and its +2 oxidation is ferrous, and valence shell with configuration $2s^2, 2p^4$ is oxygen. So,



The compound formed is ferrous oxide.

iv. Write the two important characteristics of the block type that the element Y belongs to.

→ Element Y belongs to block p. Their characteristics are:

i. Their last valence shell electron occupies the p orbital.

ii. This group includes metals, non-metals and metalloids.

3. An element X lies in group 2 & period 4.

i. Identify the element and write its electronic configuration.

→ The element X is calcium (Ca).

ii. Predict its block type & write two important characteristics of the same block.

→ As the valence electron occupy the S block the element belongs to S-block. The properties of the block are:

i. The elements have ~~valency~~ 1 or 2 valence e^- .

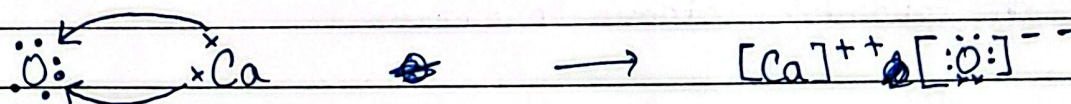
ii. They are highly electropositive and can form strong ionic compounds with non-metals.

iii. Which compound is formed when this element combines with element Y having valence shell electronic configuration $2s^2, 2p^4$ & which bond is formed between them?

→ When the element combines with element Y having valence shell electronic configuration $2s^2, 2p^4$ i.e. Oxygen. So the compound formed is calcium oxide (CaO). An electrovalent or ionic bond is formed between them.

iv. Write the Lewis-dot-structure of compound formed between X & Y.

→ Calcium oxide.



4. Answer the following questions related to Modern periodic table.

i) Write short notes on general features of modern periodic table.

→ The modern periodic table was developed by Henry Moseley which in which the elements are arranged on the basis of atomic number of elements. The table consists of 7 periods and 18 groups, the characteristics repeat after certain elements. Elements are placed in a group and periods according to the valence electrons, shell and properties. The table is divided into s, p, d and f blocks, metals are placed in s-block, non-metals ~~on the~~ in the p-block and metalloids are placed between them.

ii) Write 4 advantages of modern periodic table.

→ Some advantages of the modern periodic table are as follows:

1. It is based on atomic number, which is a fundamental property of elements.
2. Position of isotopes is explained correctly.
3. Position of lanthanides and actinides have been fixed.
4. Metals and non-metals have been separated.

iii) Predict the block type, period number and group number of following atomic numbers.

→

Atomic number	block type	Period number	Group number
7	p	2	VIA (15)
12	s	3	IIA (2)
16	p	3	VIA (16)
24	d	4	6
26	d	4	8
20	s	4	2
29	d	4	11

5. Give Reasons.

- a) Which group elements have highest electron affinity & why?
 → The group VII A (17) elements of the periodic table have the highest electron affinity as they only require $1e^-$ to become stable, they have strong effective nuclear charges.
- b) Alkaline earth metals have almost zero electron affinity. Why?
 → Alkaline earth metals have almost zero electron affinity as they have a complete s-orbital. They do not accept e^- as it would push them to high energy orbital.
- c) Which compound is formed more easily among NaCl and KCl?
 Give reason in terms of ionization energy of Na and K.
 → The compound KCl is formed more easily than NaCl as K has lower ionization energy as its valence e^- is farther than the nucleus in comparison to sodium, allowing KCl to be formed easily compared to NaCl.

d. Polarity of Hydrogen halides decreases from HF to HBr.
 Give your reasons on the basis of electron negativity of halogens.

→ Polarity of hydrogen halides decreases from HF to HBr as the electronegativity of halogens decreases down the group i.e. $F > Cl > Br$.

Since, F is most electronegative the difference in electronegativity is maximum in HF and then decreases down the group.

6. Consider one element X with atomic number 19 & another element Y with valence shell configuration $3s^2 4p^4$; ($3s^2 3p^4$).

a. Identify X & Y with their electronic configuration:

→ The element X is potassium and element Y is Sulphur.
 So,

$X \rightarrow 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1$

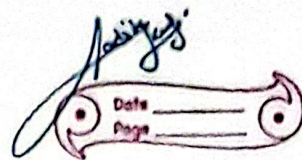
$Y \rightarrow 1s^2, 2s^2, 2p^6, 3s^2, 3p^4$

b) Which type of ions are formed by X & Y? Compare the size of ions formed with their parent atoms respectively.

→

→ Cation: K^+ is formed by element X, it is smaller than its parent atom K as when K donates an e^- it gains octet state and also loses valence shell, increasing the effective nuclear charge.

→ Anion: S^{2-} is formed by element Y, it is larger than parent atom S as ~~an~~ ~~shell~~ $2e^-$ are added the repulsion between e^- increase, making it expand.



c) Are the ions of X & Y isoelectronic? why? compare their sizes.

→ Yes, K^+ and S^{2-} are isoelectronic as both the ions have 18 electrons.

However, S^{2-} is larger than K^+ as S^{2-} has only 16 protons meanwhile K^+ has 19 protons. The protons attract the electrons more in K^+ than S^{2-} , making it smaller in size.

7. Give reasons:

a) Nitrogen is bigger than oxygen.

→ Nitrogen is bigger than oxygen as oxygen has higher ~~nuclear charge~~ effective nuclear charge, it increases along a group, due to ~~to~~ higher effective nuclear charge the atom of oxygen shrinks more and its atomic size is less than nitrogen.

b) Sodium is bigger than magnesium but smaller than potassium.

→ Sodium is bigger than magnesium but smaller than potassium as:

Sodium and magnesium lie in the same ~~group~~ ^{period}, due to higher effective nuclear charge ~~mag~~ Mg is smaller than Na.

However, Sodium and potassium lie in the same ~~per~~ ^{group} and K has one valence shell more than Na, making it larger in size than ~~Na~~ K.

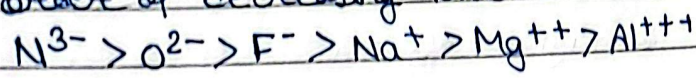
Ferric ion is smaller than ferrous ion though these are ions of the same atom.

Although ferrous and ferric ions are the ions of the same atom ferric ion is smaller than ferrous atom as both have same number of protons but ferric has lesser e^- causing it to shrink and become smaller than ferrous ion.

Chloride ion is bigger than chlorine atom.

Chlorine gains an electron to form chloride atom. Due to the electron the repulsion between the electrons increase, causing the chloride atom to expand and become larger than chlorine atom.

The ions in the following isoelectronic series are arranged in order of decreasing ionic size. Give reasons.



In an isoelectronic series the number of electrons are equal but the number of protons remain same. Inⁱⁿ isoelectronic atoms series, ionic radius decreases as nuclear charge increases because the same number of electrons are attracted more strongly by a greater positive charge.

Multiple Choice Questions

1. What is the atomic number of element of period 3 & group 17 of the periodic table?
 (a) 10 (b) 4 (c) ~~17~~ (d) 21

2. Where would you locate the element with electronic configuration 2, 8 in the modern periodic table?
 a) Group 8 b) Group 2 (c) ~~Group 18~~ d) Group 10

3. Which of these belong to the same period?

Element	A	B	C
Atomic no.	2	10	5

 (a) A, B (b) ~~B, C~~ (c) C, A (d) A, B, & C

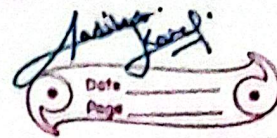
4. Name the neutral atom in the periodic table which has same number of electron as K^+ & Cl^- .
 (a) Helium (b) ~~Argon~~ (c) Neon (d) Krypton

5. An element 'A' belongs to the second period and 16th group of the periodic table find out the valency of 'A'.
 (a) Valency = 6 (b) ~~Valency = 2~~ (c) Valency = 1 (d) Valency = 8

6. Fourteen elements after actinium are called:
 (a) ~~Lanthanides~~ (b) ~~f-block elements~~ (c) ~~d-block elements~~ (d) ~~A~~

7. Moseley's periodic classification is based on:
 (a) Atomic size (b) Atomic mass (c) ~~Atomic number~~ d) ionization energy

8. Coinage metal belongs to
 (a) s-block (b) p-block (c) d-block (d) none
9. The electronic configuration of Al^{3+} is
 (a) 2, 8, 3 (b) 2, 8, 8 (c) 2, 8 (d) 2, 8, 8
10. The most electronegative element in the periodic table is
 (a) Oxygen (b) Nitrogen (c) Fluorine (d) Chlorine
11. Which pair of atomic numbers belongs to s-block?
 (a) 9, 7 (b) 7, 15 (c) 6, 12 (d) 3, 12
12. Which of the following is the metalloid?
 a) Al ~~As~~ → c) ~~Co~~ Co (d) As
 (b) Zn
13. Which of the following is not a representative element?
 (a) K (b) N (c) Cl (d) Fe
14. The group number & period number especially of an element with atomic number 8 is:
 (a) 6, 2 (b) 16, 2 (c) 6, 8 (d) 16, 4
15. What is the other name of group 18 element?
 (a) Noble gases (b) Alkali metal (c) Alkaline earth metal
 (d) Halogens
16. Which of the following element is most electropositive element?
 (a) $[Ne]3s^1$ (b) $[Ne]3s^2$ (c) ~~$3s^2$~~ (d)
 (e) $[Ar]4s^1$ (f) $[Ar]4s^2$



17. An element has an atomic number of 15 with which of the following elements will it show similar chemical properties:
 (a) Be (4) (b) Ne (10) ~~(c) N (7)~~ (d) O (8)
18. In the combined test of N & S, blood-red colour compound is formed. Which one of the following is its molecular formula:
 (a) NaCNS ~~(b) Fe(CNS)₃~~ (c) KCNS (d) NaCN
19. The correct order of first ionization energy among following elements: Be, B, C, N & O is
~~(a) B < Be < C < O < N~~ (b) B < Be < C < N < O
 (c) Be < B < C < N < O (d) Be < B < C < N < O
20. The electron affinity of inert gases is:
~~(a) Zero~~ (b) High (c) Negative (d) positive.
21. Ca^{++} ion is isoelectronic with
 (a) Mg^{++} (b) Na^+ ~~(c) Ar~~ (d) Kr
22. Which of the following ion has the smallest size?
 (a) Na^+ (b) Mg^{++} (c) F ~~(d) Al^{3+}~~
23. Higher value of electron affinity is:
 (a) Atom will lose electron easily
~~(b) Atom will gain electron easily.~~
 (c) Atom may form di-positive ion
 (d) The reason is unknown